

Kickstarter Campaign Analysis

Springboard Academy - Capstone 3

Over the last few years, entrepreneurs have invented some truly unusual ideas, such as edible picnic cups, pirate-themed pancakes, and even a bluetooth-enabled lunchbox. To bring these types of concepts to fruition, entrepreneurs have been turning to a popular crowdsourcing platform called Kickstarter. Only ~40% of campaigns are successful, and this data essay seeks to learn why. This essay will use popular statistical and machine-learning techniques to analyze campaigns with a few key questions in mind:

- How can we categorize a potential Kickstarter campaign?
- What do successful and unsuccessful campaigns have in common?
- Can we predict whether a campaign will be successful, based on its descriptive data?

The basic idea of Kickstarter is that a candidate can present a rough outline of any potential creation on the website, state how much money they believe they would need to complete the work, and viewers can choose to support by pledging funds. There is a catch: the money is only charged if there is enough audience support to exceed the entrepreneur's stated target. Otherwise, the crowdsourcing attempt fails and the candidate is left empty handed.

This project will first engage in several analytical methods to categorize campaigns and create a predictive model. Kickstarter campaign datasets are available on the data-science competitive platform, Kaggle.

- <https://www.kaggle.com/datasets/sripaadsrinivasan/kickstarter-campaigns-dataset>
- <https://www.kaggle.com/datasets/alessiocantara/kickstarter-project>

The datasets will be wrangled to create a clean, unified dataset that can support exploratory data analysis. This work will be performed in Jupyter Notebook. Common visualization techniques, such as bar plots, scatterplots, and word clusters will be created to generate a sense for the overall nature of Kickstarter campaigns. Machine-learning techniques follow, using the popular Scikit-learn library tools such as Linear Regression. In addition, taking into account categorical data both from the raw dataset and from metadata, the essay will break the campaigns into groups using K-Means Clustering.

The final step of this project will be a presentation that includes, as a final goal, a graph visualization of the campaigns. This visualization will show limited sets of campaign clusters, their relationship to each other, and their relationship to various forms of metadata. The essay will use tools such as FalkorDB, Redis, and [Cytoscape.js](#) to create this browsable static visualization. The findings will be communicated on the author's personal web portfolio.

In summary, this essay intends to use data from Kickstarter campaigns and their unusual entrepreneurial ideas to analyze and predict campaign outcomes. Foundational data-science techniques will take place first in Jupyter Notebook using Scikit-Learn, Linear Regression, and

K-Means Clustering. The findings will be communicated in an overall presentation that will include a graph visualization of clusters of campaigns and their relationships.